

TuxShow Version 1.2.0

Complete User Manual

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1 Introduction & Getting Started

1.1 What is TuxShow?

Welcome to TuxShow! Whether you are running sound cues for a middle school play, projecting dynamic video backdrops for a high school musical, or teaching students the fundamentals of projection mapping, TuxShow provides a professional feature set in a cost-effective, easy-to-learn package. It bridges the gap between basic slideshow software and highly complex, expensive industry-standard tools.

1.2 System Requirements & Launch

Because TuxShow relies on live video rendering and WebGL for effects, we recommend the following minimum specifications:

- **OS:** Ubuntu / Debian, or Raspberry Pi OS (optimized for Raspberry Pi 4 and 5).
- **Processor:** Intel Core i5 / AMD Ryzen 5, or Raspberry Pi 4/5 ARM processors.
- **RAM:** 8GB minimum (16GB recommended for heavy video playback).
- **Graphics:** A dedicated GPU is recommended for multiple 1080p outputs or complex projection mapping.

Installation: To start the application, download the newest installation package for your operating system directly from the official repository at <https://github.com/shakacb-lgtm/TuxShow>. Run the setup and launch the app—no complex licensing dongles are required.

1.3 Workspaces and “Pack Workspace”

At its core, your show is saved as a **TSW (TuxShow Workspace)** file. This text-based file is incredibly small and easily shared, saving the order of your cues, timing parameters, mapping coordinates, and absolute file paths pointing to your media assets.

The “Pack Workspace” Feature:

If you move your .TSW file to a different machine, the absolute media links might break. To make your show fully portable, use the packing feature.

1. In the top Header, click the **Pack** button.
2. Choose a destination folder (like an external USB drive).
3. TuxShow will create a `/media/` subfolder, copy every active video, image, and audio asset into it, and generate a new `packed_show.TSW` file that uses relative paths.
4. You can plug this USB drive into any other computer running TuxShow and run your show immediately.

2 The User Interface Overview

The dark-themed interface reduces eye strain in dimly lit tech booths and is divided into five functional areas.

- **Header Toolbar (Top):** Your global command center. Here you will find standard buttons for New, Load, Save, and Pack. You can also control grid sizing, toggle your Map Surface

tools, and click the green "Open Projector Screens" button to instantly send your output to connected external monitors.

- **Cue List (Left Panel):** This is the script for your show. Cues execute linearly from top to bottom. You can drag and drop media to reorder it, group items into folders, and renumber them. At the bottom, you will find your main playback controls: the large green **GO** button (fires the selected cue), **PAUSE**, and the red **PANIC (!)** button (an emergency stop that instantly fades out and stops all active cues).
- **Stage Preview (Top Right):** This live 2D canvas shows you exactly what is being sent to the projectors. You can toggle the "Map Surface" button to view the "Edit Corner Pins" overlay, allowing you to warp the main output grid.
- **Inspector (Bottom Right):** This dynamically changes to show the properties of your currently selected cue(s). *New in 1.2.0: You can now select multiple cues in your list to batch-edit properties simultaneously!* Set your Cue Number, Cue Name, Output Routing, and Geometry here.
- **Status Bar (Bottom):** Provides real-time network and hardware diagnostics. It displays your Host IP, the active port for incoming Open Sound Control (OSC) and MIDI Show Control (MSC) messages, and the virtual WebRTC stream URLs.

3 Cue Types & Capabilities

The Inspector allows you to change a cue into several dynamic types to build your show:

Cue Category	Available Cue Types & Descriptions
Media Cues	Video & Audio: High-performance, synchronized playback of visual and auditory files. Image: Static scenery, lower thirds, or announcements.
Live Capture	Camera: Pulls live video feeds directly onto the stage canvas. Supports local webcams, RTSP streams, and the new wireless WebRTC Mobile Camera protocol.
Generative	Text/Title: Renders custom typography with drop shadows and font controls on the fly. Timer: Creates live countdowns or count-ups, perfect for stage managers or intermission clocks.
Timeline Modifiers	Pause/Stop: Instantly halts or freezes a targeted media cue. GoTo: Jumps the playhead to a specific cue number (or a random range). Counter: Sets how many times a jump is allowed to execute, perfect for loops.
Hardware & Network	OSC / MSC: Sends outbound network packets to trigger changes in external sound consoles or lighting boards.
Logic & Automation	Time: Automates a cue based on your computer's real-world clock, down to the second. Conditional (If/Then): Branches the timeline based on active cues or incoming OSC sensor data. Group: Organizes your show into structural blocks. Can be set to "Fire All" simultaneously or "Fire First" sequentially.

4 Media Properties & Playback Rules

Understanding TuxShow's Unified Timing Architecture ensures your show runs precisely.

- **Triggers:** Decide how overlapping media is handled. Choose between overlapping layers or hard-stopping previous media when a new cue fires.
- **Timing:**
 - **Fade In:** How long (in seconds) the cue takes to appear or reach full volume.
 - **Duration:** How long the cue runs.
 - **Fade Out:** How long the cue takes to disappear or fade to silence.
 - **Follow Action:** Sets whether the engine should wait for an operator's manual GO or Auto-Follow directly into the next cue.
- **Transitions:** Overlay dynamic wipes on active video outputs. Choose from Iris-in, Wipe-Up, Ripple, Star, Curtain, and Wind effects.
- **Audio Controls:** Adjust output volume from 0% to 100%. Use the **Media Sync Offset** to align tracks perfectly: enter a positive value to skip forward into the track, or a negative value to delay the audio start time.

5 Projection Mapping, Geometry & Effects

TuxShow allows you to warp and correct media to perfectly align with physical stage sets or unconventional projection surfaces.

- **Master Grid:** New in 1.2.0, you can slice your global output canvas into a mesh ranging from 1x1 up to 4x4, allowing you to globally warp your entire stage output to fit complex physical sets.
- **Per-Cue Geometry:** You can manipulate individual cues directly in the Inspector:
 - **Scale X/Y & Position:** Adjust size (keeping aspect ratio) and move the center anchor around the screen using percentages.
 - **Crop Bounds & Outline:** Trim unwanted edges directly in the software, and add a custom-colored vector stroke border.
 - **Perspective Warp:** Use four blue corner pins in the Stage Preview to independently perform an Affine Transform on the specific cue, stretching it to match a quad shape (e.g., placing video inside a physical picture frame on set).
- **Visual Effects:**
 - **Color Correction:** Apply hardware-accelerated HSB (Hue, Saturation, Brightness) filters to match the mood of the scene.
 - **Chroma Key:** A real-time green screen removal tool. Use the Similarity and Smoothness sliders to isolate live actors or special effects footage without pre-rendering transparency.
 - **Masking:** Draw a custom multi-point polygon shape directly over a preview of your media to cut away parts of the image (perfect for circular set pieces or arches).

6 Networking & Broadcasting (TuxShow Ecosystem)

Your show can exist across multiple screens and devices using TuxShow's robust routing and WebRTC capabilities.

6.1 Projector Routing

Inside the Inspector, set your Output Routing. You can send a cue to "All Displays", restrict it to a specific hardware monitor (e.g., Display 2 for a stage projector), or route it exclusively to the virtual WebRTC broadcast.

6.2 Virtual WebRTC Output

When TuxShow starts, it registers a web stream on your local IP address. Turn this feature ON to broadcast the live stage canvas over your local network.

- **Web Viewer PWA:** Teachers or student crew members can type `https://<IP>:8554/display1` into their phone or laptop browser. This allows them to watch the live stage projection in real-time, perfect for remote design critiques or overflow viewing areas.
- **Mobile Camera PWA:** Have a student open `https://<IP>:8554/camera` on their smartphone. By tapping "Go Live", their phone instantly becomes a wireless, roving camera feed that you can drop directly onto the stage using a Live Capture cue!

- **Dedicated Receiver Mode:** For distributed theaters, you can spawn a borderless, auto-reconnecting receiver window on a remote networked computer, acting as a permanent, wireless digital sign or secondary confidence monitor.

7 Appendices & Technical Troubleshooting

7.1 Appendix A: OSC Command Reference

TuxShow features a robust, low-latency Open Sound Control (OSC) engine. By default, it listens for incoming UDP packets on Port 53000. Use the following syntax to control the engine from third-party software or hardware controllers:

1. Global Transport & Show Control

OSC Address Path	Argument Type	Description
/tuxshow/go	None	Triggers the currently selected cue(s).
/tuxshow/stop	None	Hard-stops all currently active cues immediately.
/tuxshow/pause	None	Pauses timeline timers and freezes playing media.
/tuxshow/resume	None	Unpauses the timeline and resumes media playback.
/tuxshow/panic	None	Soft stop; gracefully fades out cues using programmed fade times.

2. Targeted Cue Actions

Use the dynamic wildcard format `/tuxshow/cue/[cue_number]/[action]` to bypass active selection.

OSC Address Path	Argument Type	Description
/tuxshow/cue/[cue_number]/start	None	Remotely fires the targeted cue.
/tuxshow/cue/[cue_number]/stop	None	Soft-stops the targeted cue.
/tuxshow/cue/[cue_number]/volume	Float (0.0-1.0)	Updates the volume parameter in real time.
/tuxshow/cue/[cue_number]/opacity	Float (0.0-1.0)	Dynamically adjusts the visual layer opacity.

7.2 Appendix B: Keyboard Shortcuts

The following mappings allow for rapid, mouse-free theatrical operation:

Category	Shortcut (Linux / Pi)	Intended Action
Playback	Spacebar	GO (Triggers the selected cue).
Playback	Escape	PANIC (Fades out and stops all active cues instantly).
Playback	P	Toggle global Play/Pause.
Navigation	Up / Down Arrow	Select the previous/next cue in the list.
Navigation	Ctrl + F	Focus on the Cue Search bar.
System	Ctrl + S	Save current Workspace.
System	Ctrl + Shift + P	Open the "Pack Workspace" dialog.
Projection	M	Toggle Map Surface mode.

7.3 Appendix C: Theatrical & Technical Glossary

- **Affine Transform:** A geometric rendering method that mathematically stretches, rotates, and translates a 2D visual asset to fit custom 4-point coordinates without bending straight lines.
- **Follow Action:** A setting that dictates if TuxShow will "Wait for GO" or "Auto-Follow" instantly into the next cue.
- **MIDI Show Control (MSC):** An industry-standard extension of MIDI that allows lighting consoles and media servers to communicate cue triggers.
- **TSW File:** The proprietary, lightweight JSON file containing structural settings, mapping coordinates, and absolute file paths for TuxShow Workspaces.

7.4 Appendix D: Technical Troubleshooting

Running live video effects requires structured troubleshooting. Here are solutions for common issues on Ubuntu, Debian, and Raspberry Pi setups:

- **The Projector Display is Blank:**
 - Ensure you clicked the bright green **Open Projector Screens** button.
 - In your OS Display Settings, verify the desktop is set to **Join Displays** (Extend Desktop) rather than mirrored.
- **Video Playback is Stuttering:**
 - Encode all video files as 1080p H.264 (MP4) with a bitrate under 15 Mbps to avoid overwhelming the system.
 - If using Raspberry Pi hardware, try to limit active perspective warps to a maximum of 3 simultaneous video layers.
- **OSC or MSC Commands are Not Triggering:**
 - **Firewall (Linux):** Run `sudo ufw allow 53000/udp` in your terminal to allow incoming OSC traffic.
 - **MIDI Permissions (Linux):** If using a USB-MIDI interface, your user account must be in the audio group. Run `sudo usermod -a -G audio $USER` and log back in.
- **"Media Offline" Warnings:**

- This means broken file paths because the .TSW file was moved without its assets. Go back to the original computer, click the **Pack** button, and move the newly generated bundled folder to your drive.